

Amazing

Fact and Science Fiction

stories

JANUARY

35¢

MAC I

PAWN OF THE BLACK FLEET
A New Novel by Mark Clifton

TOWERS OF TITAN
by Ben Bova

THE MARS SNOOPER



Amazing

Fact and Science Fiction stories

JANUARY

35¢

MAC I

PAWN OF THE BLACK FLEET
A New Novel by Mark Clifton

TOWERS OF TITAN
by Ben Bova

THE MARS SNOOPER



The Project Gutenberg eBook of The towers of Titan

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: The towers of Titan

Author: Ben Bova

Illustrator: Ed Emshwiller

Release date: December 5, 2023 [eBook #72339]

Language: English

Original publication: New York, NY: Ziff-Davis Publishing Company, 1961

Other information and formats: www.gutenberg.org/ebooks/72339

Credits: Greg Weeks, Mary Meehan and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK THE TOWERS
OF TITAN ***

Across the frozen cliffs they loomed—the unbelievably ancient towers with the unimaginable engines deep inside them still pouring out their endless power. Dr. Sidney Lee, back from living death, vowed to find the secret of ...

The TOWERS of TITAN

By BEN BOVA

Illustrated by EMSH

**[Transcriber's Note: This etext was produced from
Amazing Stories January 1962.**

**Extensive research did not uncover any evidence that
the U.S. copyright on this publication was renewed.]**

The landing port at Titan had not changed much in five years.

The ship settled down on the scarred blast shield, beside the same trio of squat square buildings, and quickly disgorged its scanty quota of cargo and a lone passenger into the flexible tube that linked the loading hatch with the main building.

As soon as the tube was disconnected, the ship screamed off through the murky atmosphere, seemingly glad to get away from Titan and head back to the more comfortable and settled parts of the Solar System.

The passenger, Dr. Sidney Lee, stood by the window-wall of the main building and watched the ship disappear into the dark sky. He was tall and lean, seemingly all bone and tendon, with graying dark hair and faintly haunted eyes set deep into a rough-hewn, weather-worn face. When the ship was nothing more than another star overhead, he turned and looked at the place.

Five years hasn't made any difference, he thought. The single room of the main port building was unchanged: a little grimmer, perhaps, and a little more worn. But essentially unchanged. There were the same turnstiles and inspection machines, the same processing and handling gadgets for your papers and baggage, the same (it couldn't be, but they *looked* the same) two bored techs sitting at the far end of the room, unwilling to lift an eyebrow unless specifically commanded to do so.

Lee walked over to the papers processor and pushed his credentials into its slot. After a few wheezes and clanks its green light flickered on and the papers fell into the "return" bin properly stamped and approved. At the same time, his lone bag slid along a conveyor belt and onto the pickup table.

With his credentials back inside his jacket and the bag in hand, Lee looked toward the two clerks. They were studiously avoiding his eyes, searching intently through some schedules that they kept on hand for just such emergencies.

"Sid! Hi!"

He turned at the sound of her voice.

"I'm sorry to be late," she said, hurrying across the big empty room, "but we never know when the Ancient Mariner is going to arrive. It's not a matter of whether it'll be on schedule or not ... just a question of guessing how late it's going to be."

He smiled at her. "Hello, Elaine. It's good to see you again."

She hadn't changed either, and this time he was glad of it. She was still slim and young, her hair a reddish gold, her eyes gray-green. She was dressed with typical casualness: comfortable boots, dark slacks and sweater that outlined her trim figure, and a light green scarf for a touch of color. Outwardly, at least, she seemed cheerful.

"Come on," she said, "I've got a buggy in the parking area. I wanted to get a few more of the old gang to come out and meet you, but there's not many of them left, and they're all pretty busy...." Her voice trailed off.

"I didn't expect a brass band and a key to the city," Lee said. Then he added, "You're pretty gay for a female scientist," he said.

"I'm always gay when I meet old friends again."

He said nothing.

"I wish you'd cheer up," Elaine coaxed.

"I will; give me time."

They entered the parking building and got into a bubble-top car. Elaine gunned it to life, and they slid out of the near-empty parking area, through the pressure doors, and into Titan's unbreathable atmosphere.

"Have you been here straight through since I left?" he asked.

"No. I spent about eighteen months on Venus, slushing through the swamps in search of ruins that couldn't possibly have survived a century in that climate."

"And?"

"That's it," she said, shrugging. "Something ventured but nothing found. So I asked to be returned here."

"It's got you, too, hasn't it?"

Her face became serious for the first time. "Certainly it's got me. It's got all of us. Do you think we'd stay out here otherwise?"

"Anything new turned up?"

Elaine shook her head. "Nothing you haven't seen in the reports. Which means nothing, really."

He lapsed into silence and watched the frozen landscape slide by as the car raced along Titan's only highway. They crossed a bleak, frozen plain, bluish-white in the dim twilight from the distant Sun. The stars twinkling in the dark sky overhead made the barren scene look even colder. The road climbed across a row of hills, and as they made a turn around the highest bluff, Saturn came into view.

No matter how many times Lee had seen the planet, it had always thrilled him. Now, five years later, it was still an experience. Three times larger than the full moon as seen from Earth, daubed with brilliant yellow, red and orange stripes, and circled about its middle by the impossible-looking rings, Saturn hung fat and low on the horizon, casting shadows stronger than the Sun's.

"It's a compensation, isn't it?" Elaine said.

Soon they were down on the plain again, but now it was a shattered, broken expanse of jagged rock and ice. A greenish methane cloud drifted over the face of Saturn, and Lee finally turned his eyes away.

"You can see the towers from here," Elaine reminded him.

"I know," he said. He could not make out any detail, but there they were, just as they had been for—how long? Ten thousand years? A hundred thousand? Five towers jutting straight up from the bleak plain, clustered around a central, taller tower.

"Is the machinery still running?" he asked, pointlessly.

"Of course."

"There was some talk a year or so back about trying to stop it."

She shook her head. "They wouldn't dare."

The machine had been discovered more than ten years earlier, when the first Earthmen landed on Titan. Saturn's largest satellite was devoid of life, a world of dark and cold, of hydrogen atmosphere and methane clouds, of ammonia seas and ice mountains.

And there in the midst of it all stood the machine: a brazenly unconcealed cluster of mammoth buildings, with its five stately towers surmounted by the soaring central sixth. And within, row upon row of unexplained machinery, fully automated, operating continuously in perfect order.

Alien.

The discoverers soon concluded that the machine was unbelievably old, older than the Egyptian pyramids, perhaps even older than the Martian canals. And it was running smoothly. For untold centuries, for uncounted millennia, it had continued to operate efficiently, tended only by automatic machines.

A clear challenge to the space-rovers from Earth. Who made this machine? How does it work? Why is it here? What is it doing?

As soon as its discovery was made known, the machine was visited by a steady stream of Earthmen—physicists, archeologists, engineers of a thousand different specialties, and soldiers, politicians, men who were now forced to believe the inevitable. The machine was photographed, x-rayed, blueprinted, analyzed spectroscopically, philosophically, even theologically.

Who built it? How does it work? Why is it here?

No answers.

Dr. Sidney Lee, an anthropologist who had made a name for himself by unraveling the history of the ruins on Mars, arrived on Titan full of optimism and enthusiasm. Twenty months later he was taken from Titan to a psychomedical center on Earth—completely irrational and suffering from man's oldest dread: the unknown.

Returning to the underground center that had grown over the years near the machine, to house the living and working quarters of the tiny scientific community on Titan, was something like returning home for Dr. Lee. Someone had seen to it that he got his old quarters back again. Most of the people he had known from five years ago had gone elsewhere, but a few remained.

Lee spent his first few days renewing acquaintances and meeting the new men and women. He was surprised at their youth, until he tried to recall how he must have looked and acted when he first arrived on Titan.

"Makes you realize how time takes its toll, regardless of geriatrics," he said to Dr. Kimball Bennett. Official director of the center, Bennett had called Lee into his office for a chat.

"Come on now," Bennett scoffed, "you're talking like a man of ninety. Why, you won't need geriatrics for at least another month."

They both laughed. Bennett was a shy-looking, slender astrophysicist who spoke softly, never seemed to exert himself, and yet commanded the unabashed admiration of every member of the center.

"All right," Lee said. "You didn't call me to discuss my failing years. What's on your mind?"

"Oh, I just thought it's about time you got to work. You've been loafing around for a week now. We can't afford to feed you free forever, you know."

"No, I guess you can't," Lee agreed, smiling.

Bennett leaned back in his chair and studied Lee for a long moment. "I won't ask you why you wanted to come back. But I was delighted when I saw the paperwork with your name on it. Want to know why?"

"Now I am curious."

"I want to leave Titan. I've been heading this operation for too many years, now. I want out. And I really can't leave until I have a top-notch man to run this little show. You're my replacement."

"As director?"

"Yes."

"Not me," Lee said, shaking his head. "I couldn't."

"Why not?"

"Why? Hell, Kim, you saw them carry me out of here five years ago. How do you know the same thing won't happen again? How do I know it?"

A trace of a smile flickered across Bennett's face. "Look, the fact that you returned to Titan—to this center and to that infernal machine out there—well, that's proof enough to me that you've licked whatever it was that caused your breakdown."

"Maybe you're satisfied," Lee countered, "but how about the rest of the staff? How will they feel about having a reconditioned neurotic heading the show?"

Bennett's smile broke into an open grin. "Self-pity is a terrible thing. Do you know what those kids think of you? You're Dr. Sidney Lee, the foremost xeno-anthropologist of the human race. You're the man who deciphered the Martian Script, who uncovered the ruins on Tau Ceti, who did the definitive studies on the cave man cultures on Sirius and Vega. Your troubles here on Titan are just a six-month incident in the middle of a dazzling career. Haven't you noticed the deference with which they've been treating you? You're a big man on this campus."

"I don't know...."

"It won't be a lifetime job," Bennett coaxed. "In a couple of years some of the young squirts around here will have acquired enough poise and self-control to run the show. Then you can go on to something else."

Lee got up from his chair and paced slowly to the bookshelf that lined one wall of the office. "Why don't you stay on for another year or so, and then turn it over to one of the youngsters?"

Bennett wordlessly extended his arms over the desk. His hands were trembling, almost imperceptibly, but trembling.

Lee stared at the hands. "You too?"

Bennett placed his hands palms down on the desk. "Do you think you're the only one who worries about an alien race who can build a machine that we can't understand after ten years of investigation and study?"

The official transfer took place the following day, shortly after breakfast. Bennett called a meeting of the six department heads and announced that he would leave Titan on the next ship. It came as no surprise.

"From here on, I'm just an interested observer," he said. "Dr. Lee is in charge." He paused for a moment, then went on, "I thought this would be a good time to review what's been going on most recently, and where we stand."

Lee took an old pipe out of his jacket pocket and filled it while he watched them try to decide who would talk first. The six department heads all looked young and eager, he thought. Elaine was among them, of course, as head of the archeology group.

After a bit of finger-pointing and head-shaking, Dr. Richards took the floor. Head of the physics section, he had one of those open, clear-eyed, crew-cut faces that would look young even after his hair turned gray.

"It just so happens," he began, "that we finished a study yesterday that may be of some slight significance." From the size of his grin, Lee judged that the physicist was making a weak attempt to underplay his speech.

Richards walked to the viewscreen at the far end of the room and turned a dial. The screen flickered for a moment, then showed a chart.

"You remember from our last meeting," he said, "that our group finally succeeded in reaching the generating unit that powers all the machinery out there. Uh, Dr. Lee, this is something we've been working on for more than a year. The power unit is buried in the sublevel of the main building out there, and it's damned difficult to get to it without tearing out other machines. We finally wormed a man down there about a month ago."

Richards turned back to the rest of the group. "This chart shows what we've been able to learn about the power unit. Which isn't much. There's no input to it. No batteries, no solar cells, nothing. There's a fuel tank—at least, we think it's a fuel tank—that's sunk inside a cryogenic magnetic coil."

Elaine spoke up. "You told us that last time. Have you been able to get into the tank?"

Richards shook his head. "Not unless we break up the coil, which we don't dare try. It would probably mean destroying the power unit and stopping all the machinery. And we can't do *that* until we're certain of what the machinery's doing."

"We all know that," said Dr. Kulaki, a wiry Polynesian who headed the electronics group.

"Yes. Well, we did make a very elegant experiment," Richards continued, "a variation of Cavendish's experiment to obtain the gravitational constant, in the Eighteenth Century...."

"Spare the history," Dr. Kurtzman said, half-smiling.

"Okay, okay," Richards said. "We determined the mass of the fuel tank, and therefore of the fuel in it. The tank contains a degenerate gas...."

"What?"

"A degenerate gas," Richards repeated. "The stuff must weigh several tons per spoonful."

Dr. Petchkovich, the astronomer, frowned puzzledly. "Wait a minute. Degenerate gases are found only in the cores of certain types of stars ... are you saying that this magnetic field they've put around the fuel tank is strong enough to create a pressure similar to the weight of a star?"

"Take a look at the chart," Richards said, pointing to the viewscreen. "If you know of any other type of substance with a density like that, I'll eat it."

There was a brief conversation, then Dr. Bennett said, "All right, Pat. Is that all?"

"Hell no," Richards said, his cat-like grin returning. "There's more."

He tapped a button on the viewscreen control panel, and another chart came up on the screen.

"We know the power output of the generator. That was simply obtained, since the one unit powers all the machinery in the buildings out there. So we calculated all the known methods for obtaining power from a degenerate gas, and checked them against the amount of fuel the system has used up so far...."

Lee interrupted. "How do you know how much fuel has been used if you can't get into the tank?"

Richards' grin broadened. "Oh, that's easy, Dr. Lee. We know how much a degenerate gas should weigh, per unit volume. We know the size of the tank, and therefore how much it can hold, when full. So we can estimate how full the tank is simply by measuring its mass and comparing it to the mass it would have if it were full."

"Doesn't the chemical composition of the gas have any effect on the mass? Wouldn't uranium be heavier than hydrogen?"

"Yes, but not much. A factor of a hundred. You'll see in a minute that it's not enough to matter at this stage of the game."

Lee nodded and the physicist went on. "Well, anyway, you can check out our math in the report we'll issue later this month. But it turns out that the only possible energy source for this gadget is total annihilation of the gas particles into energy."

"Total annihilation? How?"

"That's a good question. I don't think we'll be able to answer it until we can start taking the damned machine apart." He flicked a new graph on the screen. "But, we can calculate how long the thing has been running, on the basis of the fuel it's used up, and the energy rates we've assumed...."

A slow wave of astonishment crept through the small room as, one by one, they grasped the significance of the curving lines on the graph.

"That's right," Richards said. "Unless our rough calculations are completely off the beam, which I doubt, the damned thing has been operating continuously for something like ten-to-the-fifth or ten-to-the-sixth Earth years."

A hundred thousand to a million years.

The rest of the meeting was quiet and orderly. They were all subdued by Richards' report. *It's been running continuously for a million years*, Lee kept thinking. *A million years*.

He listened automatically as the other department heads made their reports.

Ray Kurtzman was first. His report was actually a combined discussion of the work that his engineering group, and Dr. Kulaki's electronics people, had been jointly undertaking. They had tried to determine (for the *n*th time) just what sort of power was being beamed by the antennas atop the machine's towers. No luck. The machinery was using power, the antennas were broadcasting *something*, but whatever it was could not be detected by any instrument the Kurtzman and Kulaki had applied to the problem.

Elaine gave a routine report on the latest digging expeditions that had been sent out. No artifacts, no foundations, no remains of any sort. Blank.

Dr. Childe, a short, sharp-voiced mathematician, gave his report on his department's analysis of the wave patterns that were being sent out by the antennas. The patterns had been deduced from the fluctuations in energy consumption by the antenna equipment. Childe reported another blank. The patterns were completely random.

"It's foolish to call them patterns at all," Childe complained. "They look more like the ramblings of an idiot than anything produced by intelligence."

Finally, Petchkovich reported on the latest astronomical studies. The antennas were tracking an empty section of space between the Sun and the planet Mercury. Careful observations had shown no noticeable effect in the widespread area where the antennas were focused.

"It is my belief," he concluded, "that the antennas were originally focused on one of the planets, but have since become disoriented in some way, and are now well off-target."

Kurtzman huffed. "Not very likely. If they—whoever they are—could make this whole damned set of buildings full of machinery to operate continuously

for a million years, do you think they'd slip up on where the antennas are pointing?"

"If it's a million years we're talking about," Petchkovich answered slowly, "then the chances for errors are simply that much greater."

Lee decided it was time for him to step in. "Have there been any other attempts to date the buildings or the machines? Radioactive decay rates, or something like that?"

Elaine nodded. "It's been tried at least once or twice a year, every year. Nothing conclusive has ever been established. The buildings are obviously very old ... but million years...."

They all lapsed into silence.

Lee took a deep breath and began once more. "This million-year-business throws a new light on the whole subject. Up to now, I think, we've all been looking at this machine on a day-to-day basis. I mean, we've been examining what it's doing *now*. It might be worth our while to sit back a little and try to extrapolate the behavior we've observed back over a million years."

"Could you be a bit more explicit?" Richards asked.

"I'll try. It's mostly a matter of viewpoint, I think. Let's start looking at this machine as something that's been in operation for a million years. Let's admit to ourselves that what we see today is only a very small slice of the whole picture. Dr. Petchkovich, you're an astronomer; you're accustomed to studying a star for a few months and coming up with a story that covers perhaps billions of years. Right?"

"Yes, but this is entirely different...."

"I know. I know," Lee said. "But the line of attack isn't so different. For instance: Dr. Childe, suppose you send your wave patterns to the Orbital Computation Center at Earth and have the Big Brain work them over. Do you think it might come up with something useful?"

Childe shrugged elaborately. "Maybe. If we try to extrapolate the sample patterns we have now over a million years, maybe something will show up."

"It'll take months to get the Big Brain's time," Elaine said. "They're waiting in line all the way back to Mars for it."

"What are months compared to the time we've already spent?" Bennett countered.

"Or to a million years?" Lee added.

Kulaki suddenly started to bob up and down in his chair. "Say, we might be on the track of something here," he said. "If those circuits have been in continuous operation for a million years ... we could learn an awful lot about reliability...."

Lee nodded in agreement. "We have a lot to learn, that's true enough." He cleared his throat nervously. "There's one more thing, I am about to publish a paper ... it's a sort of a general paper, but it has some bearing on the work going on here. I wonder if you'd be good enough to be a tryout audience for me?"

They sat back to listen.

Lee gave only the basic outline of his paper. He discussed his findings among the ruins on Mars and on the lone planet circling the star Tau Ceti; and he drew some conclusions from his investigations of the primitive human cultures found on the planetary systems of Sirius and Vega.

First, both Sirius and Vega have both been long known to be comparatively young stars. Astrophysical evidence compiled from the Twentieth Century onward, and finally geophysical data from the planets themselves, showed that Sirius and Vega—and their planets—were considerably less than a billion years old. By contrast, the Solar System was known to be at least five billion years old.

Now, the development of life takes time. It took close to three billion years for life to make its first appearance on Earth. Another two billion years of evolution were necessary before man arose.

"If the planets circling Sirius and Vega are less than one billion years old," Lee stated, "then the human populations of those planets—no matter how primitive they are—*could not have originated there*. They must have come from another planetary system. The closest system that could have born life on its own is our Solar System."

Second, the ruins on Tau Ceti and on Mars were both definitely built by human beings. The plans of the buildings, their furnishings and utensils, the scant writings and pictures that survived—all were of human origin.

"Although dating the time of destruction of these buildings is very difficult," Lee went on, "we can definitely say two things: they were destroyed suddenly in some immense cataclysm; and they were destroyed at very nearly the same time, both on Mars and Tau Ceti. I hadn't realized the correlation until this morning, but the destruction might easily be dated at approximately one million years ago."

There was a stir of reaction at the mention of the time period.

Also, Lee told them, he had been able to partially translate one of the folk tales of the Vegan people.

"Stripped of its nonessentials, it tells of a cataclysmic war between the ancestors of the Vegan people and another race, mentioned only as the Others."

He paused and looked at them, sitting around the table. They seemed engrossed.

"The Vegan folk tales relate that the Others won an overwhelming victory, and smashed forever their Golden Age, when they travelled through the skies and lived in unbelievable splendor.

"I'd like to make one final point. The Vegan and Sirian peoples are hardly human, by our standards. In addition to slight physiological differences, they live in caves and eat shellfish and giant insects. They do not hunt meat-bearing animals, simply because there are not yet any meat-bearing animals on their planets. The humans on those planets did not evolve there; they are a billion years ahead of the natural evolution of their worlds."

Bennett spoke up. "In other words, they were originally a colony from Earth."

"Exactly."

"That's a lot to swallow in one sitting," Patrick said, quietly.

"I know," Lee answered. "But it fits in with what we've found here. Elaine, I'm certain that if you knew where to look, you could find on Earth the remains of a human civilization that is more than a million years old...."

She shook her head. "That's contrary to every known scrap of evidence about early man. Besides, a million years of weathering would wipe out almost anything on Earth. Wind, rain, Ice Ages, earthquakes ... it would be impossible."

"What about the Moon?" Petchkovich asked. "There should be some archeological evidence on the Moon that hasn't been completely destroyed. Maybe buried under the dust or lava flows...."

"At any rate," Lee said, "I'm convinced that there was a human culture a million years ago; that it expanded out through the Solar System and to the stars; that it met an implacably hostile alien race; and that the humans were utterly defeated. Nearly wiped out."

"And the machine out there?"

"It was built by the Others. It's alien. And hostile," Lee said. "That's why we must find out what it's doing, and how."

You could feel the change in attitude throughout the center after that conference. An invisible, but almost palpable wave of tension swept through every office, every lab, every section of the underground community. Not everyone believed Lee's theory about the Others. But they could never look at those strange buildings again without wondering. What had been a puzzling, frustrating riddle took on the attributes of a living, intelligent enemy; an enemy that mocked their efforts to understand, to control; an enemy that seemed in every way superior to the feeble powers of the men from Earth.

A few days after that conference, Dr. Bennett left Titan for good. Lee was in full command now.

He was just finishing a quick lunch at the robocafeteria when Elaine walked by and stopped at his table.

"Hi boss. I'm on my way out to the buildings. Want to come along?"

Lee punched a button at the end of the table, and the dishes slid into a wall receptacle. "No thanks," he said. "I've got some paperwork to do...."

"Sid, you're going to have to come out and face it sooner or later. Why don't you come now? There's practically nobody there."

He looked up at her. "You're not a part-time psychotech, are you?"

"No," she said, grinning. "But I wouldn't want anybody to start thinking that you're afraid to go into the buildings. And I especially don't want *you* to think that."

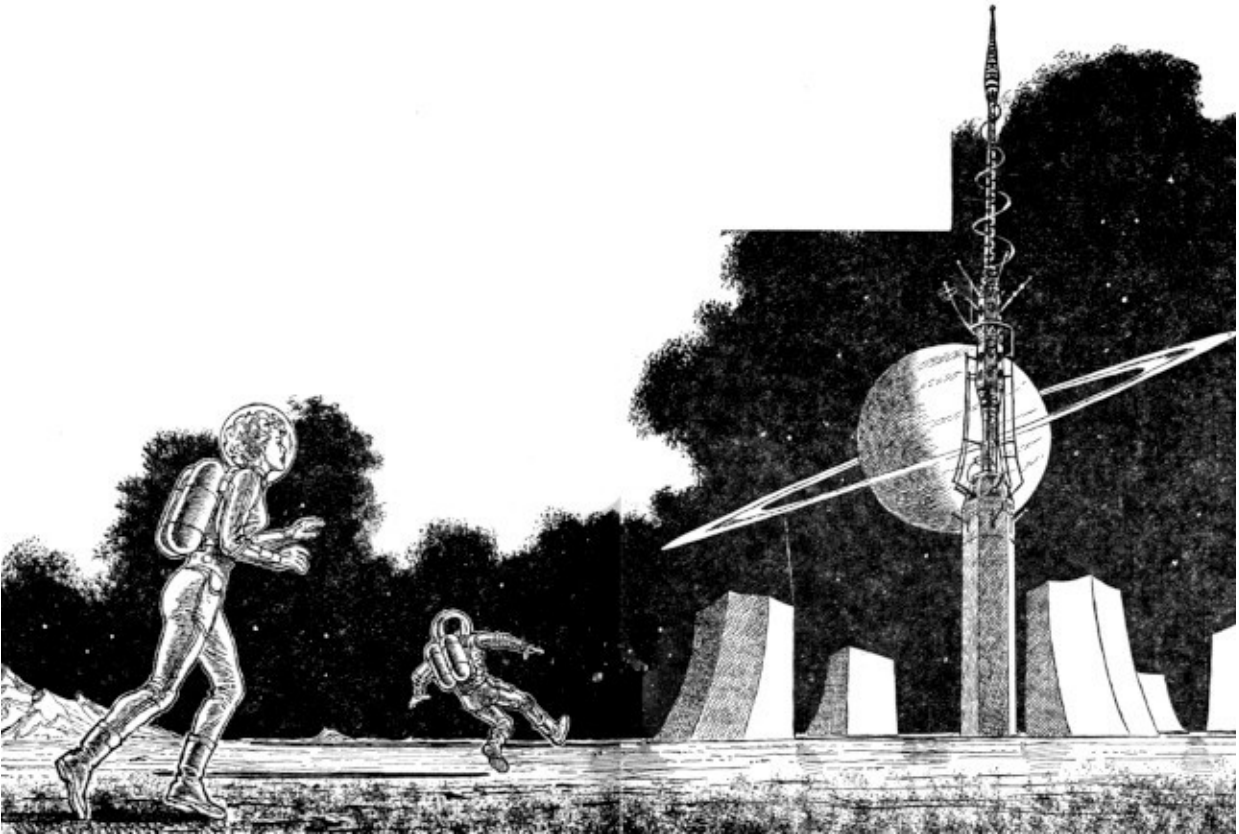
"Okay," he said, getting up from his chair. "You're right ... as usual."

They took a lift chute to the surface bubble, a dome of clear plastisteel that housed ground vehicles and miscellaneous outdoor equipment. There they squirmed into pressure suits, complete with fishbowl helmets. After checking out their oxygen tanks, heaters and radios, they were ready to go outside.

"Want to walk?" Elaine asked.

Lee nodded.

It took him a few minutes to get accustomed to the low-gravity shuffle that you must employ to walk on Titan, with its one-third Earth gravity.



They headed for the buildings, under the double shadows cast by the distant Sun and the ever-present, overpowering Saturn. The buildings loomed straight up from the dark plain, gaunt gloomy specters from a bygone age haunting this shadow world. There were five low, square featureless structures ringed around a central pentagonally-shaped tower that swept upward to a series of spires and antennas. Several doors had been cut into the outer buildings' walls by the inquiring Earthmen. Originally, the walls had been perfectly blank.

"Did you bring a torch?" Lee asked.

Elaine shook her head. "Don't need one anymore. We installed lights inside. They're tripped by a photocell as we cross through a doorway."

He could feel it coming on as they approached the buildings—the tenseness, the prickling along the spine, as if a deeply-buried memory was writhing within his mind.

Even before they entered the doorway he could sense the throbbing, beating *purposefulness* of the machines.

And then they were inside, surrounded by them, row on row, tier on tier, inhuman untiring infallible machines humming, growling, whining, filling the vast building with the rumbling power of their work. Driving, constantly driving at their unknown tasks. Along catwalks that snaked through the maze of machines, automatic maintenance vehicles scurried along, stopping here for a quick adjustment, there for replacement of a faulty part.

No matter where the two invading humans went along the twining catwalks, the maintenance vehicles avoided them. If they stopped before a machine that was to be serviced, the maintenance vehicle would hover nearby, glowering at them, waiting for them to get out of the way.

He could feel it again—the alienness, the lurking presence of an intelligence that scorned the intruders from Earth. Every nerve in his body screamed the same message: get out, get away, this thing is evil, hostile, a weapon against all mankind. In the conference room, telling them of his theory, he had inwardly wondered how much of it he himself really believed. But here, in the midst of these implacably efficient machines—he *knew*. This is the product of a cosmic hatred, the work of those who seek to destroy man, our ancient enemy, the unknown, the nameless Others.

"Are you all right?"

Elaine's voice in his earphones snapped him back to reality.

"Why, do I look green?"

She came up close enough so that their helmets nearly touched. "A little green," she said, smiling. "It gives me the creeps, too. Want to go?"

"No," he answered. "Let's see the work that's going on."

He took a firm grip on himself and went through it all, from the combined crews of archeologists and engineers tracing wiring circuits, to the handful of physicists conducting further tests on the power generator buried deep in the central building's foundation.

"It goes awfully slowly," Elaine said as they trudged back to the center.

"We're on the outside looking in," Lee said. "If we could only determine the *purpose* of the machinery, then the rest of it would come pretty easily."

They returned to the surface bubble and took off their pressure suits.

"Nearly dinner time," Elaine said, as they descended to the living area. "How about eating in my quarters?" she asked.

"Can you cook?"

"You'll find out."

A few hours later, he had decided that she definitely could cook, and that she had somehow managed to bring to Titan some of the best wine he had tasted in years.

Now the table was folded back into the wall, and they were sitting together on the couch, listening to music tapes.

"What's that?" he asked.

"Villa-Lobos, a Twentieth Century composer. *Bachianas Brasileiras*. Number eight, I think...."

They listened in silence for a few moments to the moody, restless music.

"Reminds me of topside, standing at the shore of the ammonia sea when a storm is coming up," he said.

Elaine nodded. "Yes ... the wind, and the dark, and the waves...."

Abruptly, the music snapped off.

"Oh damn!" Elaine blurted. "I thought I had the thing fixed."

"Let me take a look at it," Lee said getting up from the couch.

He tinkered with the recorder for a while, to no avail. "I'm no electronics tech, that's for sure."

Elaine shrugged. "Don't worry about it. I'll have it repaired tomorrow."

She went into the kitchenette and called back, "How about some brandy?"

"Fine," he answered.

She returned and handed him a glass.

"Ad astra," he said, as they touched glasses.

"Amen."

Elaine walked slowly away from him, swishing the liquor in her glass. "Let's go topside and see the stars," she said, suddenly.

"H'mm?"

"Up to the bubble. Come on."

So they left her one-room quarters and took the lift tube up to the bubble. It was deserted. Overhead, Saturn was low on the horizon, silhouetting the alien buildings. They turned to the other side of the sky, where it was clear and dark. Thousands of stars twinkled in the darkness; they made out Sirius and Vega, then Mars, and finally Earth—a bright blue jewel that outshone all the others.

"I guess Jupiter's on the other side of the Sun," Lee said.

Elaine nodded. "It looks so far away," she said, staring at Earth, "and so lonely, out there in all that emptiness."

"A psychotech would call that 'projection.'"

"I know. We're the lonely ones, aren't we?"

She was perfectly serious now.

"You know that Ruth and I have split up," he said.

"I heard ... that she left you."

He shrugged. "That's a matter of viewpoint. She got tired sitting Earthside while I batted around the Solar System. When I told her I was going to Vega, she called it quits."

"Couldn't she have gone with you?"

"If she had wanted to...."

"Do you still love her?" Elaine asked.

"I don't know. I don't think I know what love really is, anymore. All I know is ... on that long trip out to Vega, when I had nothing to do but sit and think, it wasn't Ruth I was thinking about. It was you."

"Oh...."

They talked a bit more, and finally he took her in his arms and kissed her and she was his, at least for a little while.

Weeks lapsed into months, and the work on Titan inched steadily along. If he stopped to think about it, Lee knew that all they were doing was scratching around the base of the problem, and making precious little headway. *The blind men and the elephant*, he told himself. But then he asked himself what else they could possibly do, and the answer was always, nothing. But the machine was still there, doing whatever it was designed to do, and he could sense the scornful laughter of its creators as he vainly tried to understand their work. Only the thought of Elaine, the sight of her, the touch of her, allowed him to keep his sense of balance.

People left Titan, baffled and confused; new people arrived—eager, full of energy, excited at the chance to tackle the unknown, undimmed—at first—by the day-to-day frustrations of trying to unlock a door that has no key.

Lee dialed his selection at the robocafeteria and waited a few moments for the tray of food to appear at the pickup table. He had spent the morning shut away in his office, searching months worth of reports for some glimmer of encouragement. There was progress, of course; there was always progress. But it was never in a direction that would take them closer to the final answer.

And, carefully tucked into the top drawer of his desk was a nasty yellow sheet that bore a querulous message from Earth: What is the status of the

project? Why are expenditures constantly climbing? Is a ninety-three-man staff really needed? When can some solid results be expected?

Lee picked an unoccupied table and sat down. As he started to eat, a quartet of young engineers, headed by Dr. Kurtzman, came in and sat at the next table.

"I still don't see why they keep digging," one of them was saying. "They'll never find anything outside the buildings."

Another countered, "Look, they've got to follow every possible angle. The only way we're ever going to understand this thing is to put together every scrap of evidence we can find until there's enough to form an understandable picture."

Kurtzman shook his head. "Not at all. This isn't going to be solved by putting pieces together, like a puzzle. There's more evidence in those buildings than we can ever hope to digest. It's not a matter of adding up clues ... this is going to be a *gestalt*. One of these days, somebody's going to get a few thousand million of his brain cells turned the right way, and he'll say, "Ah-HAH!" Then we'll have it. Until then, it's our job to keep poking around, hoping to find the right piece of information to trigger the *gestalt*."

The first one spluttered. "But that ... that's non-scientific!"

"So?" Kurtzman asked, arching his eyebrows. "Do you see science making any great strides around here? We're in over our heads. Intuition is the only thing that can save us."

"If we're that bad off, we might as well quit," the first one said. "In fact, that might be the best idea of all. Forget it and go home. Let the damned machine run for another million years, if it wants to."

Lee could not keep quiet any longer. "That would be fine, wouldn't it?" he said, turning on the surprised engineer. "Give it up and forget about it, without knowing where it came from, or what it's doing, or why."

"I didn't mean...."

"Listen to me. It's not just that the thing could be a weapon. It may not be. But we don't know. And as long as we don't know, we've got to keep trying to find out. Understand?"

"Yes, but...."

"And there's more at stake here than just an intellectual puzzle," Lee insisted. "If we turn our backs on this machine, we've turned our backs on a basic premise of all scientific thought. If we admit that we can't understand this machine, then we admit that there's an absolute limit to our ability to understand the universe. We give in to the old witch-doctor's claim that there are some things in the world that man must not tamper with. Taboo!

"The basic nature of man and science is at stake here! We've got to understand that machine. Our claim to the stars is tied up in it."

Lee looked up and saw that everyone in the dining hall was watching him. He stood up.

"Sorry; I didn't mean to get so vehement," he mumbled to the engineer. "Guess I'm a bit edgy today."

He walked quickly out of the dining area and returned to his office.

Slowly, quietly, the work went on. Dr. Petchkovich spent six weeks on Mercury, supervising at first hand the investigation of the area where the machine's antennas were focused. He returned to Titan in high excitement.

"We have definitely proved that there is a disturbance in the interplanetary magnetic field at the focal point of the machine's antennas," he announced to the department heads, when they convened in the conference room to hear his report.

"How strong a disturbance?" Dr. Kulaki asked.

Petchkovich hesitated a moment. "Well, it's only one part in a hundred thousand...."

The excitement died quickly. It was a discovery, yes. But it did not bring them any closer to understanding the machine and its purpose.

Lee was sitting at his desk, staring moodily at a graph that Dr. Childe had left with him: the results of the Orbital Computing Center's extrapolation of

the wave patterns broadcast by the machine.

The phone buzzer sounded. Without taking his eyes from the graph, Lee flicked the phone on.

"Are you busy right now?"

He looked up and saw Elaine's face on the screen. "No, not really," he answered. "Come on in."

"I'll be there in a few minutes."

As soon as she came through the door he knew that something was wrong. She was trying too hard not to look concerned.

"What's up?" he asked her.

She shrugged and sat down in front of the desk. "I just thought we might talk for a while."

"In my office?"

"It concerns official business."

"I see."

She looked at the graph. "What's that? Something new?"

Changing the subject, he thought to himself. *What's she afraid of?*

Aloud, he answered, "It's Childe's results from the Big Brain ... the machine's wave patterns extrapolated over a million years." He turned the graph around so she could see it better.

Elaine studied it for a moment, then shook her head. "Childe was right, wasn't he? It doesn't make any sense at all. No pattern, no rhyme, no reason."

"I don't know," Lee said. "There's something ... well, something strange about these waves. They don't follow a regular pattern, and yet...."

"What?"

He frowned. "They're just too damned irregular to be really random. That doesn't make much sense, does it? Well, anyway, I've got the feeling that I've seen this pattern—or lack of pattern—before. Somewhere I've seen something that looks a lot like this ... but I can't remember where."

Elaine looked at the graph again, at the multiple curves swinging back and forth across the paper, intertwining in seeming confusion. "It looks like some of the graphs I drew when I was an undergrad."

He laughed. "Maybe that's where I saw it."

There was an awkward silence. Lee got up from his seat and walked around the desk. He pulled up another chair and sat beside Elaine. "Now, what's the trouble?"

"I've been thinking," she said slowly. "I ... I've decided to transfer off Titan."

"Leave? But why?"

"For us, Sid. For both of us. Before we get so wrapped up in each other that we won't be able to break it up without really getting hurt...."

"I don't understand," he said.

She spoke calmly and softly, no tears, no hysterics. She had thought it all out very carefully. "Sid, take a look around you. We're like two castaways on a desert world. We love each other—here and now. But we won't always be here. What happens when we leave Titan? What happens when we're not faced with the loneliness and that ... that *thing* out there? Suppose we find that we don't really need each other? What then?"

"But I do need you, Elaine. I love you."

"You do now," she answered. "But how long will it last? Sid, I have to get away, at least for a little while. I need a sense of perspective."

"I see," he said, shifting his gaze away from her. "That doesn't leave me much to say, does it? All right, Elaine. Make out a transfer request and I'll sign it."

"Thanks. I ... I hope you understand." She got up and started slowly toward the door.

"I'm not so sure that you understand this yourself," he replied. "But I know this—I don't want to lose you, Elaine. If you haven't worked this out within a month or so after you leave Titan, I'll come looking for you."

She turned and, without a word, went over and kissed him. Then she quickly left the office.

Lee returned to his desk. He sat down and stared at Childe's graph again. After a few minutes, he angrily slapped it in a drawer, slammed the thing shut, and stamped out of the room.

Personal matters were soon buried in the excitement of another discovery, this time an important one. It looked promising.

Kurtzman and Kulaki finally discovered what form of energy the antennas were beaming out.

After years of trial-and-error experimentation, the engineer and electronicist asked Richards and Childe to lend a hand. With a firm theoretical and mathematical background to bolster their work, Kurtzman and Kulaki started out on a process of elimination.

They soon proved that the antennas were not broadcasting any known form of electromagnetic energy, from gamma to long-radio waves. They investigated one possibility after another, turning up a steady succession of negative answers. Negative, but answers all the same. It was time-consuming, but at least they were definitely determining which avenues were blind alleys.

Then Childe started tinkering with a hunch, and showed his paperwork to Richards. The two of them made a few suggestions to Kurtzman and Kulaki. Their problem was that their detection instruments were drawing blanks, when applied to the antennas' output. But the power input to the antennas showed they were working continuously.

Childe showed, mathematically, that their output must be an extremely weak, low-frequency form of energy. Richards agreed, and pointed out that there was only one known form of energy that fulfilled these conditions: gravity.

"Gravity waves?" Kurtzman asked, incredulously, when they told him about it.

Kulaki's mind reacted faster. "Knowing it's gravity waves, on paper, and proving it experimentally are two different things."

But they were up to the test. They had to scavenge equipment from the center's grav screen machinery, and the whole underground community was without its Earth-normal grav field for two and a half days, but when the field was returned to normal (and people stopped hopping and bouncing all through the center) Kurtzman and Kulaki had proved that the antennas were indeed beaming out gravity waves.

It's all here, Lee thought, as he read their combined report. Now we know what the machine is doing. But to what purpose? What influence does this have on Earth?

Then he put down the report and turned to another bit of paperwork that lay on his desk: Elaine's official request for a transfer. He looked through it automatically. There was a place marked *Justification for Request*. Elaine had typed simply, "Personal." Lee flipped to the last page and signed.

He leaned back in his chair and tried to let his mind float free; forget about her, forget about the buildings out there, forget about everything ... at least for a few moments. Just drift and let your mind wander where it pleases....

The automatic secretary on his desk hummed into life and announced, "Department heads' meeting starts in five minutes."

To hell with the department heads. To hell with Titan. A world of cold and dark and ice. Like Dante's Inferno. Titan is hell. A place where sinning scientists are sent. Their punishment is to stand out in the cold and try to solve the unanswerable. Forever. Stand in the icy darkness and try to understand the unknown. Once in a while you make a discovery that excites you; but when you look again, your discovery is meaningless, you know as little about the real answer as you knew before. The discoveries are just part of the punishment, part of the eternal torment, they just keep you going after the carrot on the stick, but the carrot is always out of reach....

"Department heads' meeting beginning now," the automatic voice said.

Lee cocked an eye at the tiny device. "All right," he said, getting up, "I'm going."

The meeting was strictly routine. The reaction had set in. A week ago they had all been agog with the gravity-waves discovery. Now it had become apparent that the discovery had not opened the door they were trying to get through. *We're like children, Lee thought, trying to put together a stereo transceiver from an assembly kit; all the pieces are there, but we can't get them together in the proper way.*

It was a short meeting. As they broke up, Lee saw Richards walk over to Elaine.

"I hear you're leaving us," the young physicist said. "Going Earthside?"

"Yes," Elaine answered. "For a while, at least."

Richards broke into his feline grin again. "Good. I'll be vacationing on Earth in a few weeks. Do you like to ski?"

"I haven't skied in years...."

"There's a lodge I go to in Switzerland. Really fine. And the skiing is marvelous. Even if you don't want to ski, there are mountains to climb ... and glaciers...."

Lee started toward them, thought better of it, and walked sullenly out of the room. He went to his office, sat fidgeting at his desk for a while, then called her on the intercom. She was not at her office or her quarters, so the equipment automatically paged her. When her face finally showed on the viewscreen, Lee could see that she was still in the conference room, which was now empty, except for her and Richards.

"What is it, Sid?"

"Uh ... can you come down to my office for a minute? Right now?"

She quickly covered her surprise. "Of course. I'll be right down."

Within a few minutes, she was entering his office. "What's the matter?"

"Nothing," he said, gesturing her to a chair. "I just wanted to get you out of the grips of that All-American physicist."

"What? Why, you jealous old goat!"

"Never mind. If you want to listen to propositions, please make sure I'm out of earshot."

She laughed. "Is that what you brought me down here for?"

"Yes."

"I'm flattered."

"Don't be cute."

"He's a very nice boy."

He frowned. "All right, so I'm foolish and juvenile...."

"And he's so interesting," she went on, paying no attention. "He was telling me all about glaciers. In Switzerland. Near this ski lodge. Seems the glaciers have been growing during the past few years. Something I should really be sure to see."

"Sounds fascinating," Lee grumbled.

"Oh, it is. The glaciers, that is. They were retreating, you know, until a few years ago. Now they're growing again. Goes in cycles, it seems...."

Lee suddenly stiffened. "Great God of our forefathers!"

"What now?"

Instead of answering, he scrambled out of his chair and went to the floor-to-ceiling bookcase that lined one wall of the office.

"Sid, what's the matter? Did I say something...."

But he was not listening to her. "That's it," he was muttering, "That *must* be it."

Elaine watched him paw through the shelves of books, desperately searching for something.

"Can I help?" she asked.

"No. I think ..." he snatched at a book and riffled through it. "Yes! Look at it!"

He wheeled and shoved the open book at her. She nearly dropped it as he ducked back around his desk and rummaged through a pile of papers.

"Here it is. Look at them!" he shouted, handing her a sheet of paper. "Put them side by side and look at them."

The book was opened to a graph, Elaine saw, that showed the advance and retreat of the Ice Age glaciers over the past million years on Earth. The sheet of paper was Childe's graph of the wave patterns of the machine's antennas.

Her face went pale as she looked at the two graphs, side by side. "They ... they're the same ... almost identical."

"It all fits together now," Lee said, drumming his fingers on the desktop in a restless tattoo. "The machine beams gravity waves into the interplanetary plasma between the Sun and Mercury. The effect is infinitesimal, by our short-term standards, but over a hundred thousand years or so, the cumulative effect must be enough to block off a small fraction of the Sun's radiation. The Earth goes into a deep freeze for a few millennia!"

"But why?" Elaine asked. "Why did the Others build it?"

Lee paced nervously across the room. "Think a minute. They had beaten the Earthmen in a bitter interstellar war. They had done their best to wipe out the human race. What better way to insure their victory than by subjecting our homeworld to violent climate changes? They probably thought they were guaranteeing the complete extinction of mankind."

"But the Ice Ages didn't destroy man," Elaine said.

"No. They reduced him to the level of a beast, though. Those few survivors of the interstellar war were robbed of their civilization. They had to go back to living in caves, to fighting the other animals for sheer survival. They made it, though, and re-learned what was lost, and built a new civilization."

"Then the machine failed its purpose."

"Right," Lee said. "But remember, it's only an automatic machine. Those who made it, the Others, they're still out there among the stars somewhere. They're going to be mighty upset when they find out that we're not dead."

"It's ... terrifying ... in a way."

"The important thing is that we *understand*. Now we can face them, wherever they are. They're no more intelligent than we are. We've proved that by learning what the machine is all about. They may be older, and they certainly know more tricks, but they're not on a completely higher plain of mental development."

Elaine leaned back in her seat. "Well, it's over. At last it's ended. We can leave Titan for good now."

"Leave? Not likely. Now we can finally start to tear the machinery apart and see how it works. We've just been sniffing around it so far, treating it like a museum exhibit. Now we can shut the damned thing off and start dissecting it. There's a lot to be done, a whole new technology to be learned. How does it work? Who are the Others and where did they come from? There are more questions to answer now than ever."

"That means that you won't be leaving Titan?" she asked.

"Not for a while. How about you?"

She thought for a moment, then answered, "If you're right about all this, then there should be a lot of archeological evidence awaiting discovery on the Moon. Maybe I should organize a team...."

"Then you're leaving anyway," he said.

"For a while, Sid. I'll be back in time."

"Stay off the ski slopes."

She grinned and got up from her chair. Lee kissed her, and she turned away and went to the door. She looked back to say a final word, but he was already at his desk, punching buttons on the automatic secretary:

"All right, you mechanical marvel, call the department heads to the conference room for a meeting in five minutes. And take a message for the Terran Council Science Committee—Gentlemen: It is my belief that the question of the alien machinery on Titan is now essentially solved...."

THE END

*** END OF THE PROJECT GUTENBERG EBOOK THE TOWERS OF
TITAN ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.